

# GARAGE RENOVATION

## CONCEPT DRAWING SET

Architecture · structure · electrical · data · shop air

### BASIS OF SET

Current structural model: frame-20260920.01-beam-scheme.compas.json  
Current architectural/equipment layout: cabinets.py  
Units: inches. North is up on plans unless noted.  
Member and equipment locations are taken directly from the current model.  
Electrical, Ethernet, compressed-air and dust routes are conceptual.

### DRAWING INDEX

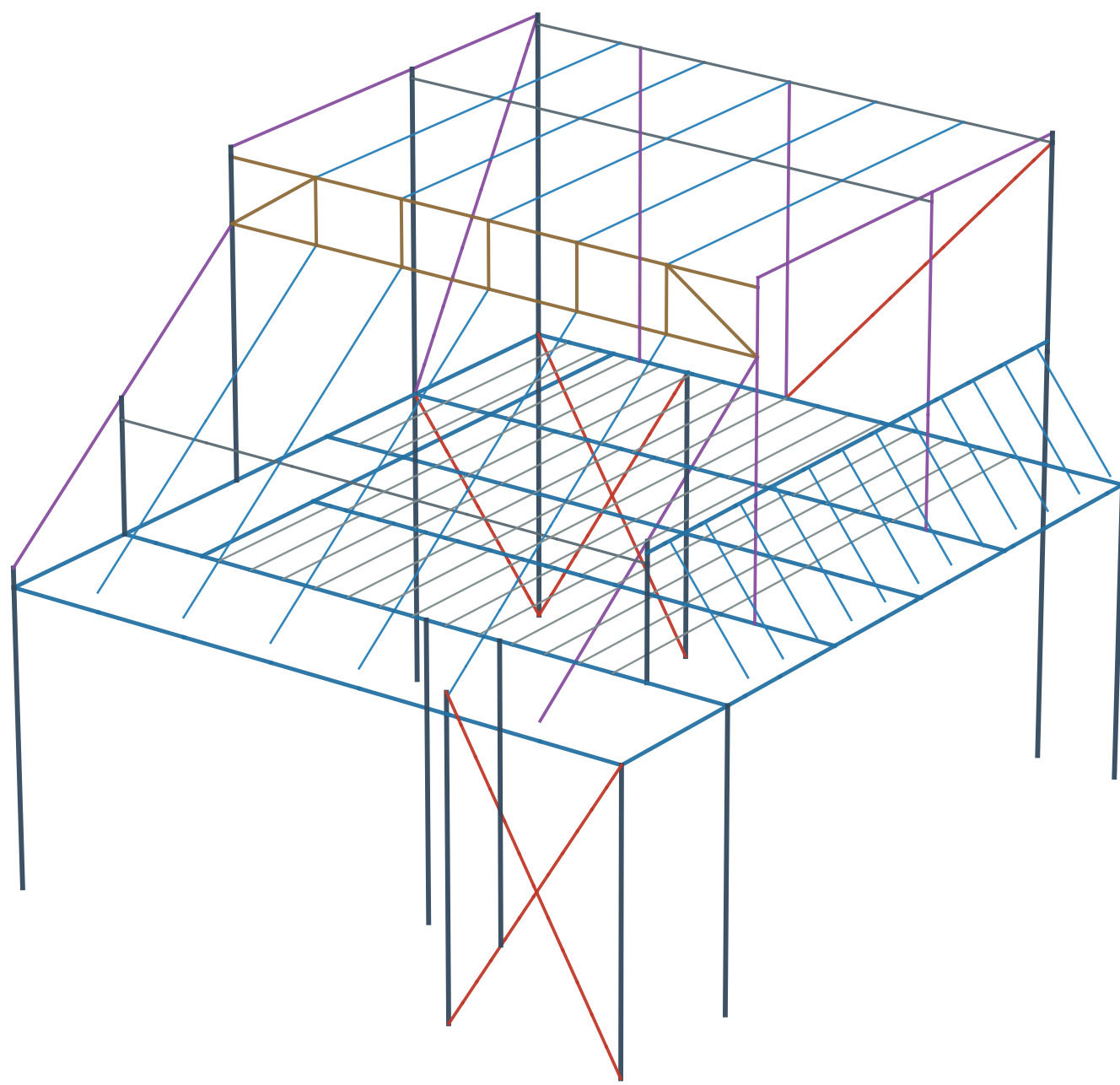
G0.01 Cover / drawing index  
A1.01 Isometric and exterior elevations  
A1.02 Loft equipment and cabinet plan  
A1.03 Ground-floor and shed equipment plan  
E1.01 Concept electrical distribution  
T1.01 Concept Ethernet distribution  
M1.01 Concept dust collection and compressed air

### USE AND LIMITATIONS

This set records design intent for coordination and discussion. It is not a permit set, fabrication package, or construction document.  
A licensed design professional must verify dimensions, loads, connections, foundations, fire/life-safety work and all MEP requirements before construction.  
Proposed service routes show endpoints and zoning; conductor, cable, pipe and duct sizes remain to be designed.

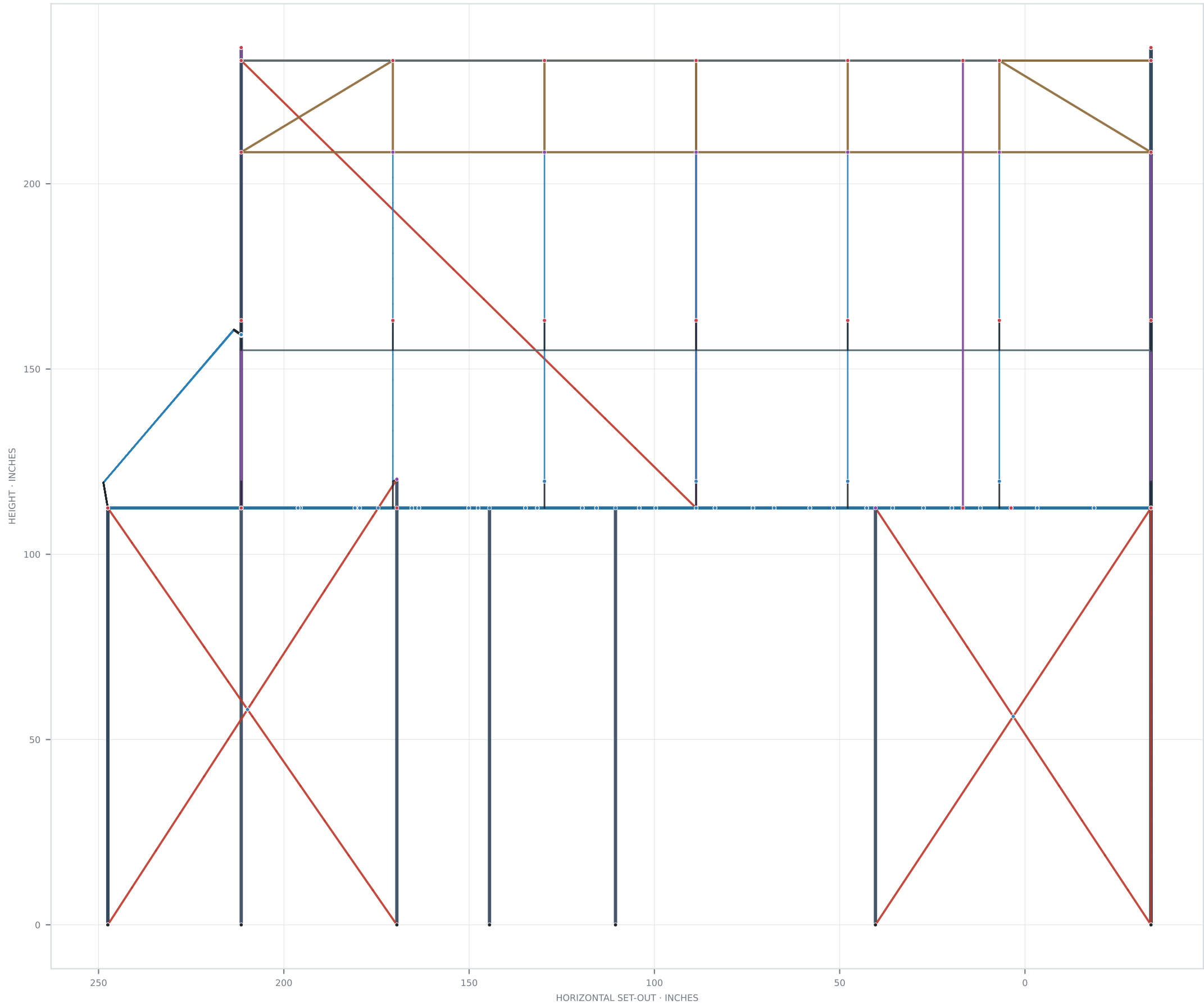
ISOMETRIC — CURRENT FRAME

SCALE: DIAGRAMMATIC



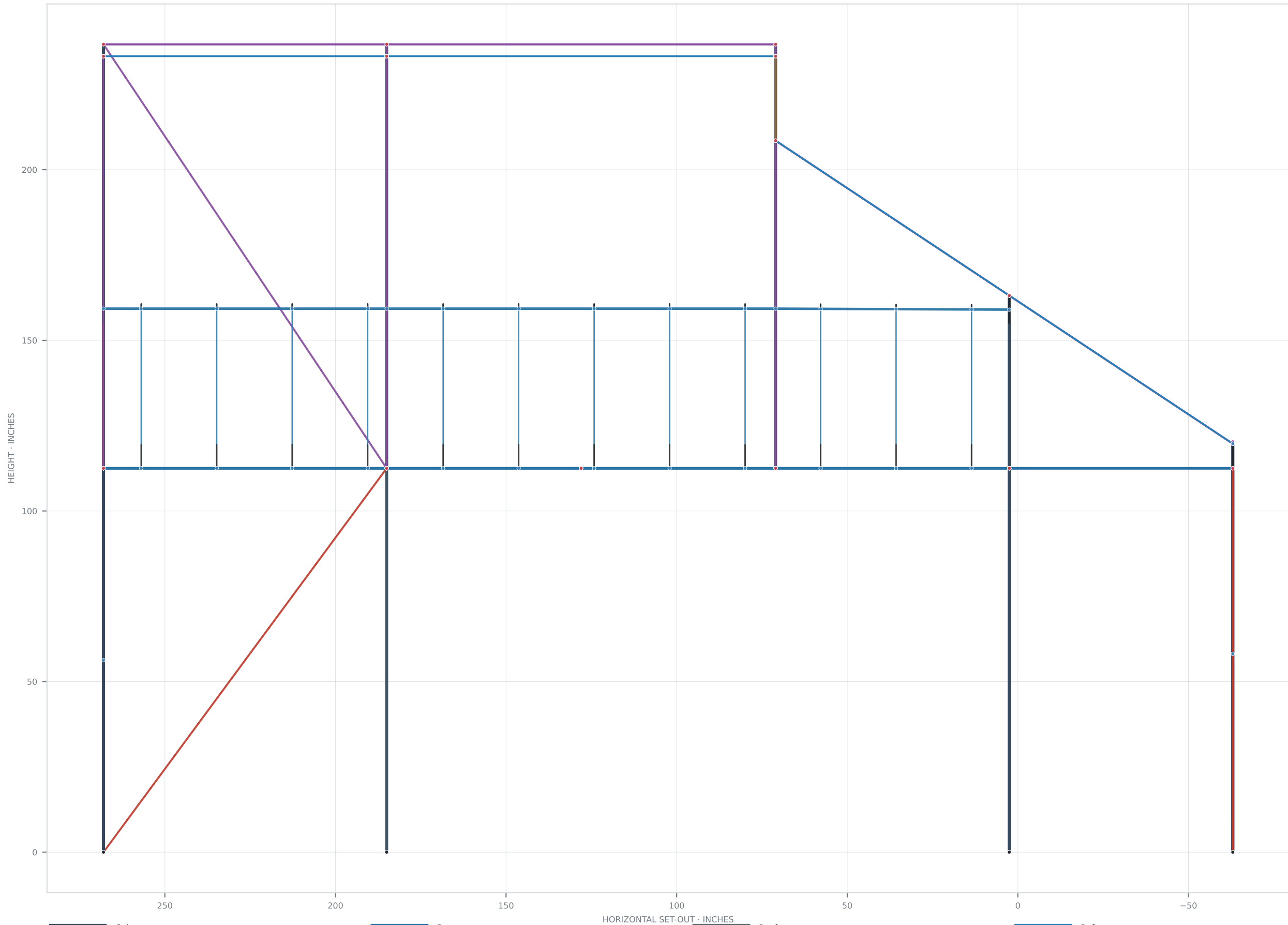
NORTH ELEVATION — LOOKING SOUTH

SCALE: DIAGRAMMATIC



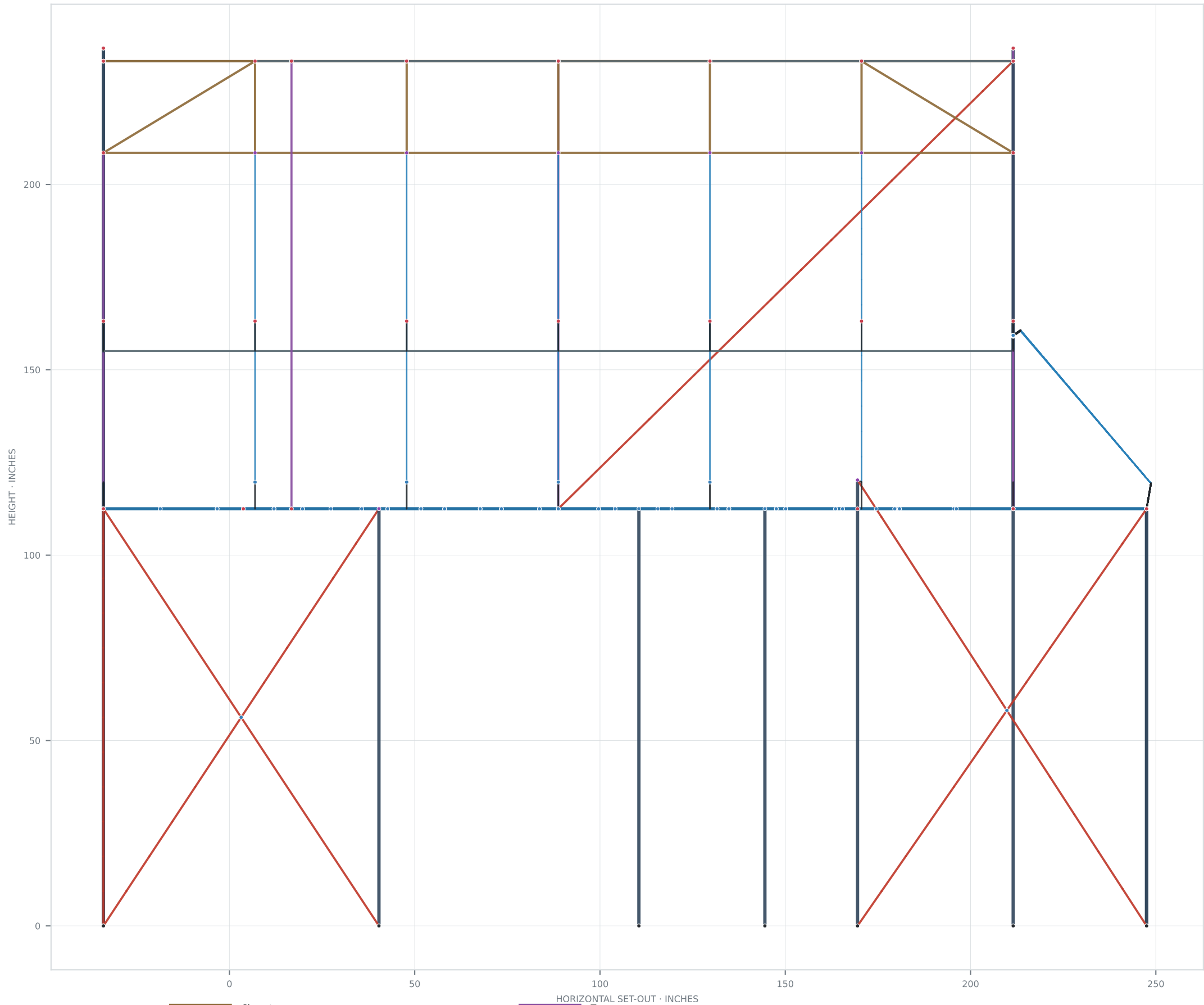
EAST ELEVATION — LOOKING WEST

SCALE: DIAGRAMMATIC



SOUTH ELEVATION — LOOKING NORTH

SCALE: DIAGRAMMATIC



COLOR KEY

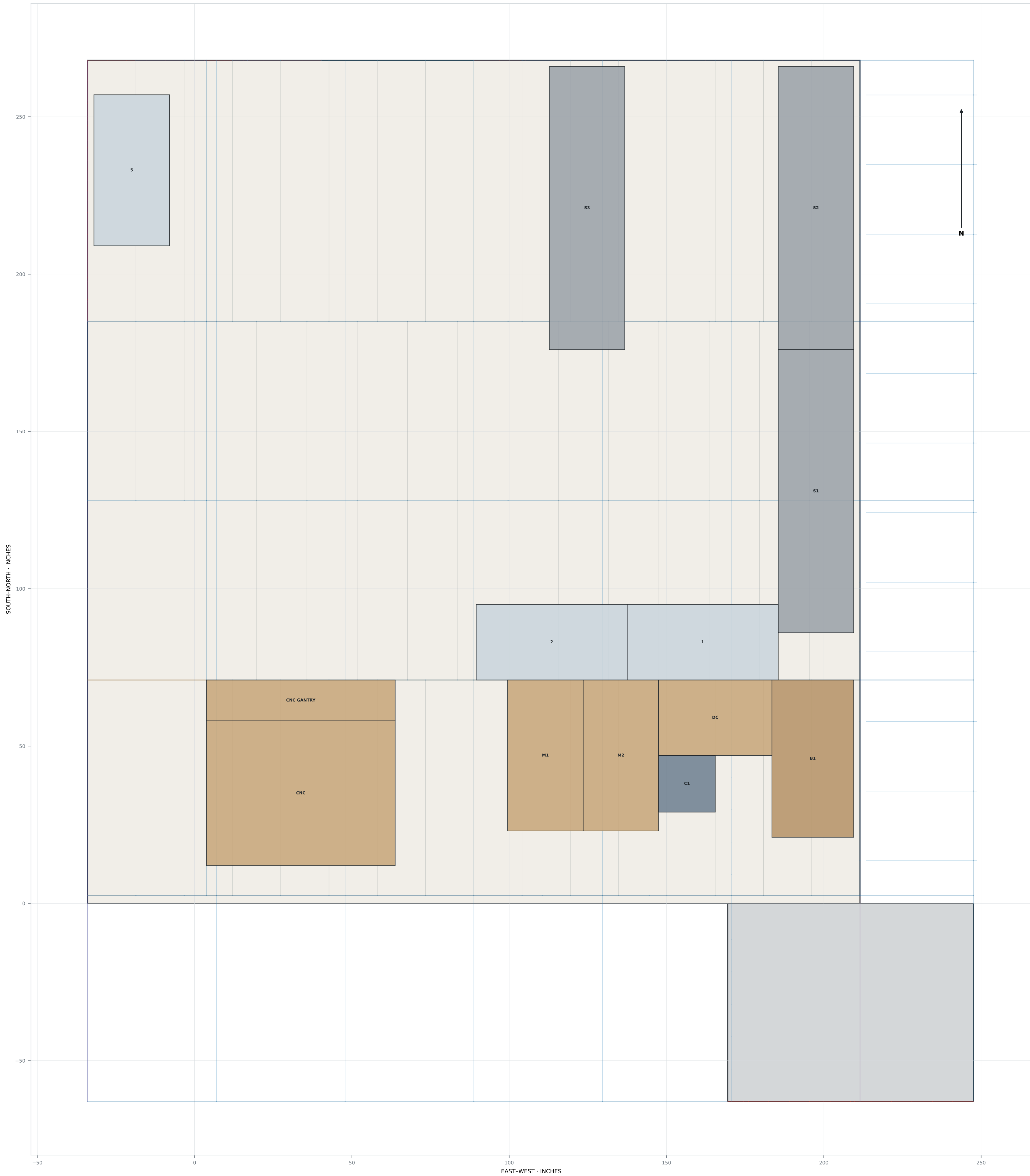
- Columns
- Beams
- Roof
- Rafters
- Joists
- Bracing
- Clerestory
- Truss

JOINTS

- Welded — moment-continuous
- Welded joint with pinned attachments
- Pinned — bolted, seated or hung
- Pinned base — anchor rods

LOFT PLAN

SCALE: DIAGRAMMATIC



LOFT EQUIPMENT

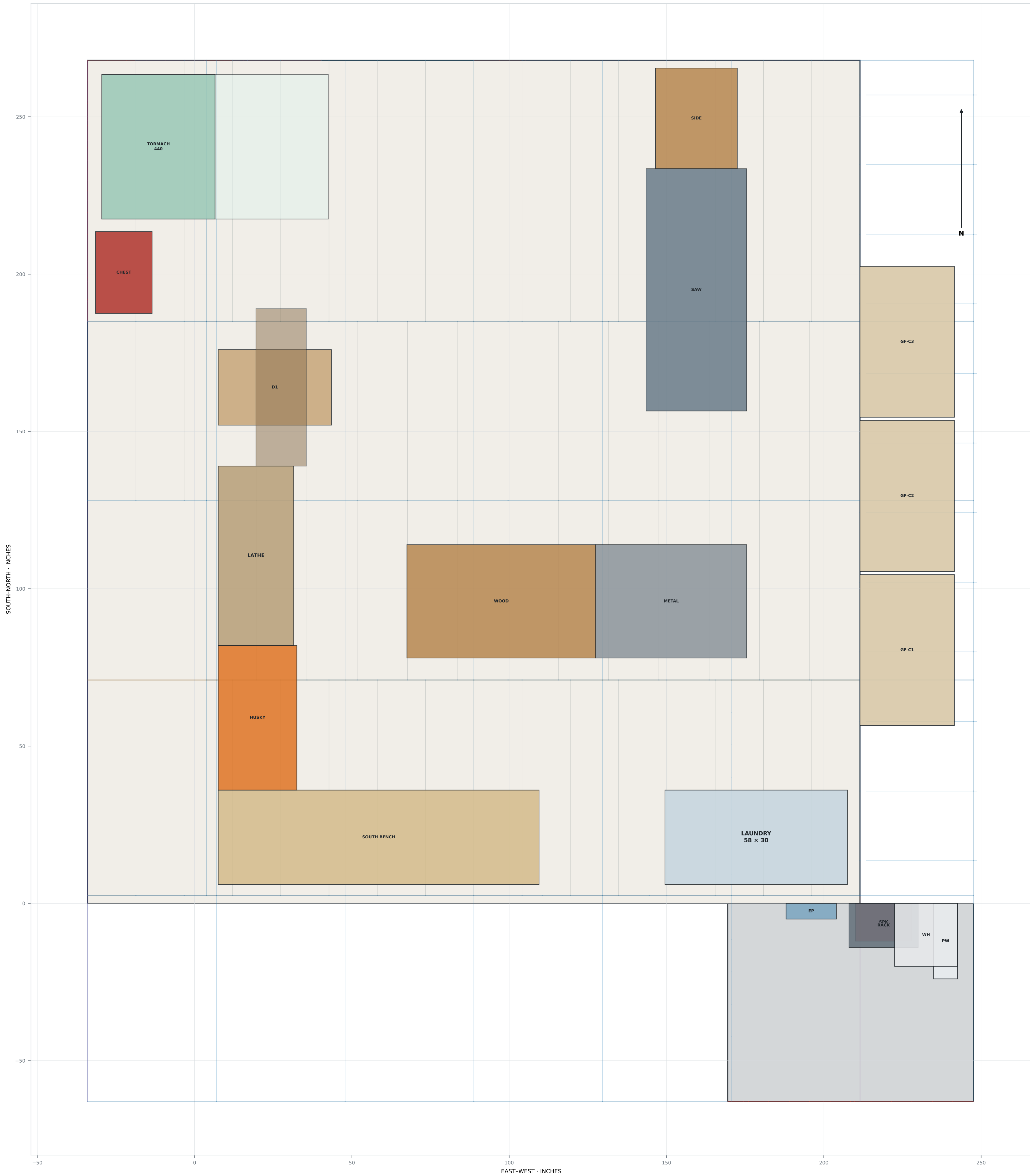
- 1 cabinet 48 × 24 × 72 in
- 2 cabinet 48 × 24 × 72 in
- 5 cabinet 24 × 48 × 64 in
- B1 crate 26 × 50 × 46 in
- M1 machine 24 × 48 × 40 in
- M2 machine 24 × 48 × 40 in
- S1 shelving 24 × 90 × 90 in
- S2 shelving 24 × 90 × 90 in
- S3 shelving 24 × 90 × 90 in
- C1 compressor 18 × 18 × 36 in
- DC machine 36 × 24 × 60 in
- CNC machine 60 × 46 × 37 in
- CNCg machine 60 × 13 × 53.25 in

COORDINATION NOTES

Cabinets and equipment are nonstructural.  
Maintain the documented walkways and access zones.  
Verify anchorage, working clearances and concentrated floor loads.  
CNC is shown in its parked position.

GROUND-FLOOR PLAN

SCALE: DIAGRAMMATIC



GROUND-FLOOR ELEMENTS

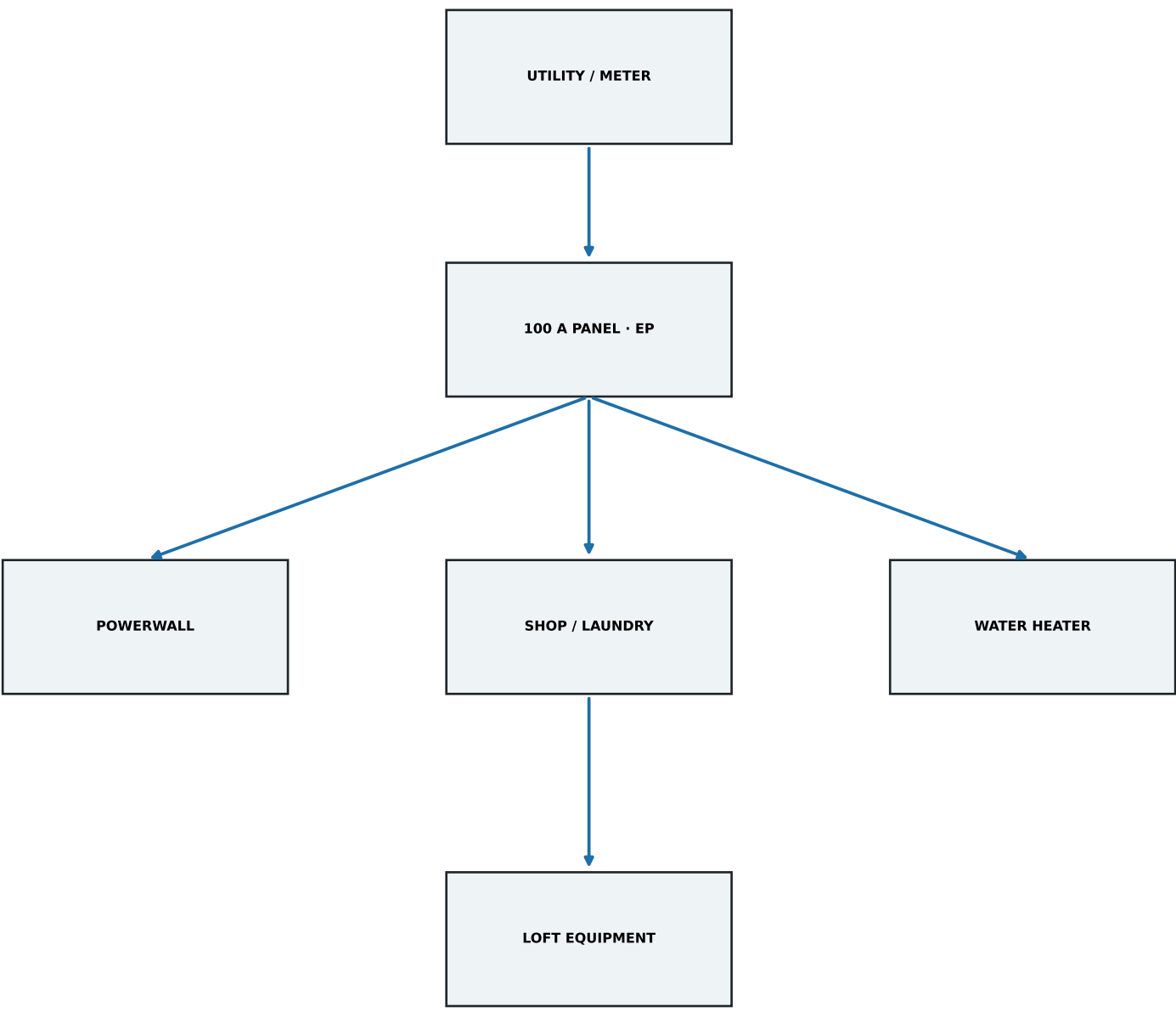
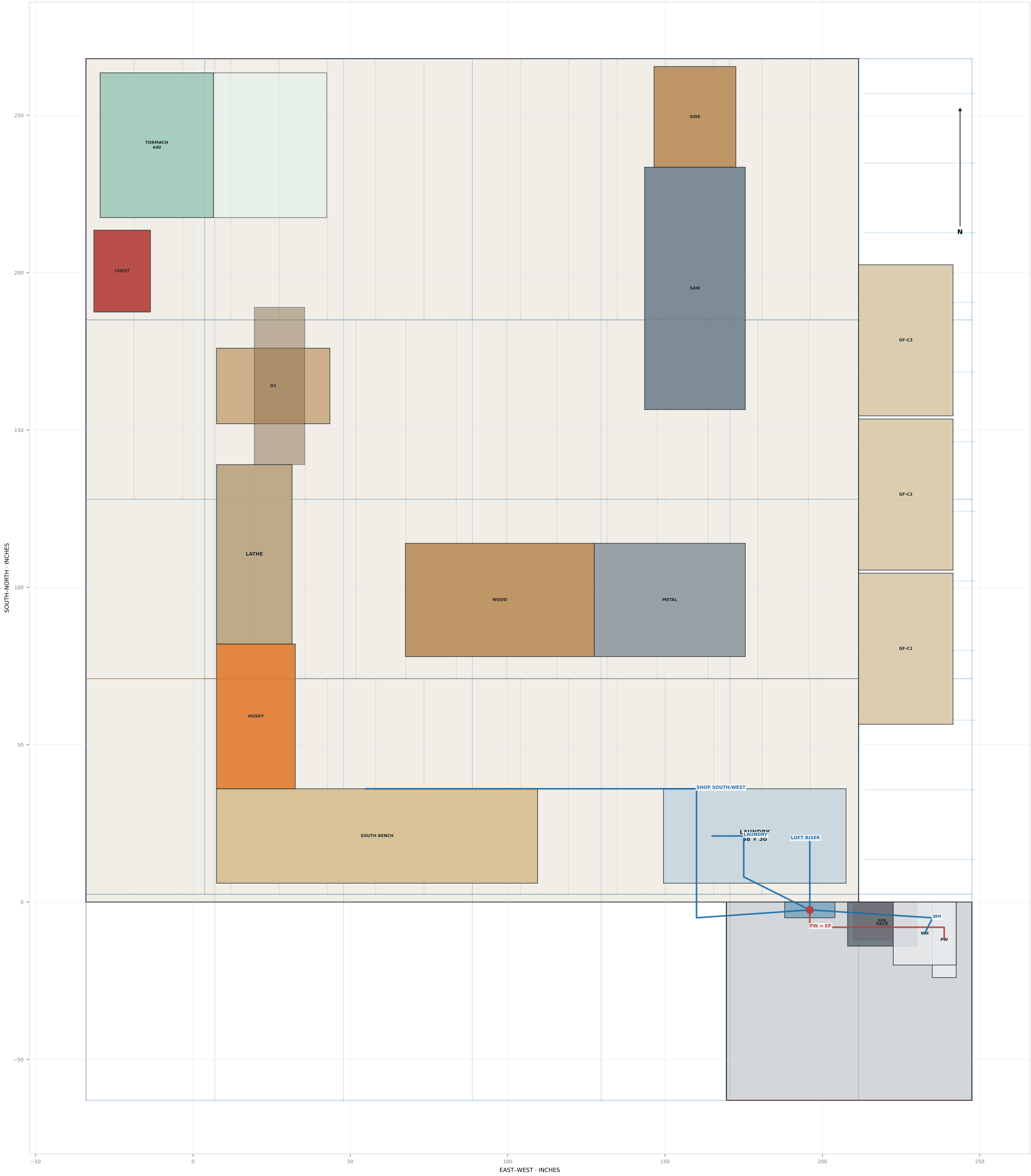
- GF-C1-C3 east built-ins
- LAUNDRY 58 x 30 x 66 in
- SOUTH BENCH 30 in deep
- CHEST Craftsman 26 x 18 x 58 in
- HUSKY rolling cabinet 46 x 25 x 36 in
- D1 RF-30 mill/drill + 50 in. table
- WOOD / METAL tables 60 x 36, 48 x 36
- SAW saw/router 32 x 77 x 38 in
- SIDE table 26 x 32 in
- LATHE 24 x 57 in
- PW Tesla Powerwall 3 - 291 lb - east wall, over the water heater
- EP 100 A electrical panel - conceptual 16 x 5 x 30 in. envelope
- SPK sprinkler equipment cabinet - conceptual 18 x 12 x 24 in. envelope
- RACK wall-mounted 19 in. rack - conceptual 12U - 14 in. deep
- WH 15 gal electric water heater - 20 in. diameter x 29 in. high

COORDINATION NOTES

Shed equipment is shown at the current cabinet-layout positions.  
Keep required electrical-panel, heater and sprinkler service clearances.  
Confirm wall construction and equipment anchorage before installation.

POWER PLAN — CONCEPT ROUTING

SCALE: DIAGRAMMATIC

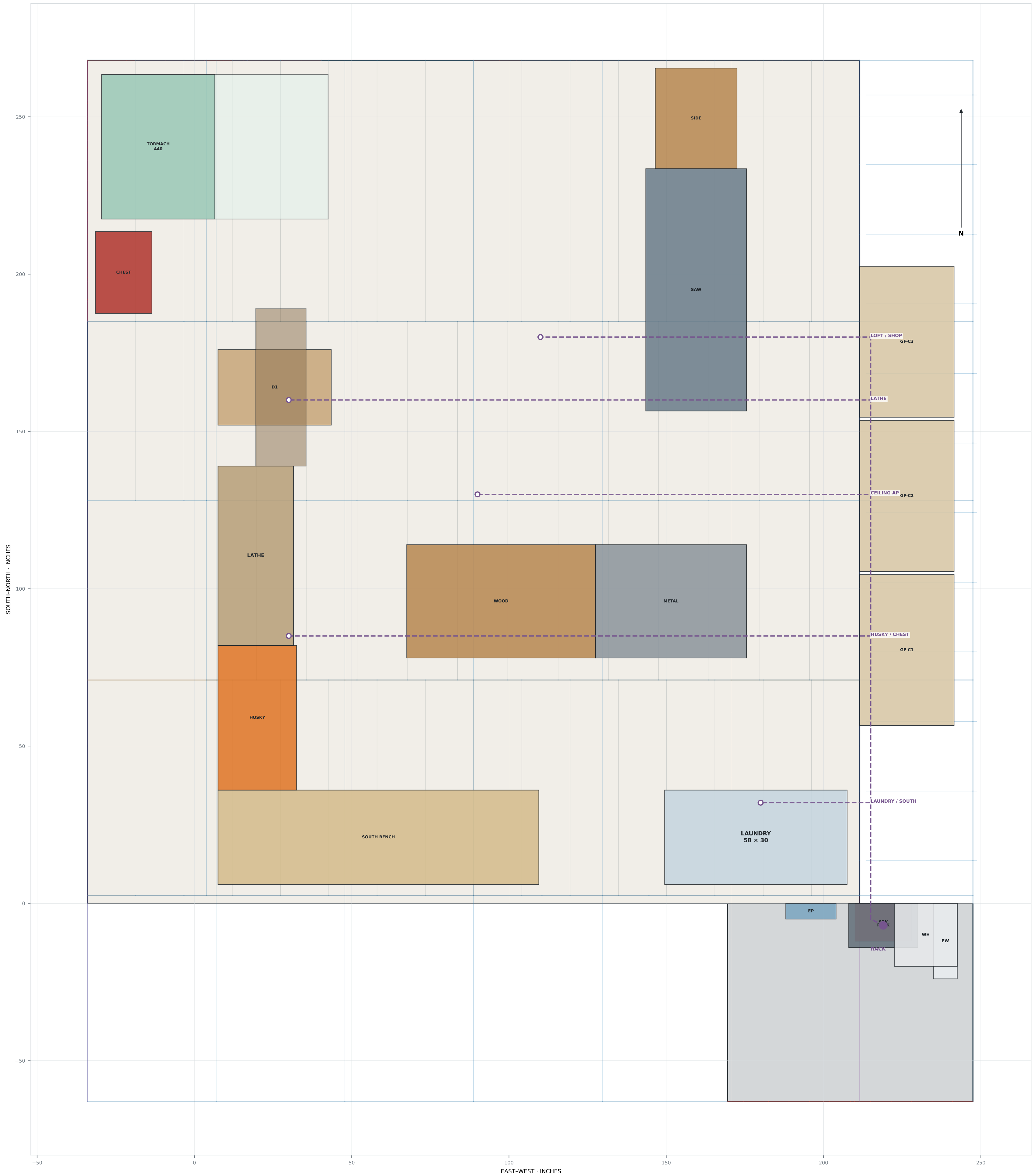


DESIGN DECISIONS REQUIRED

Confirm service point, available fault current and Powerwall topology.  
Develop load calculation and panel schedule; size all feeders and branches.  
Provide dedicated circuits as required for laundry, water heater, CNC, mill/drill, dust collector and compressor.  
Coordinate disconnects, GFCI/AFCI protection, grounding, lighting, receptacle spacing and code working clearances.  
All routes shown are conceptual home-run zones.

DATA PLAN — CONCEPT HOME RUNS

SCALE: DIAGRAMMATIC



SYSTEM CONCEPT

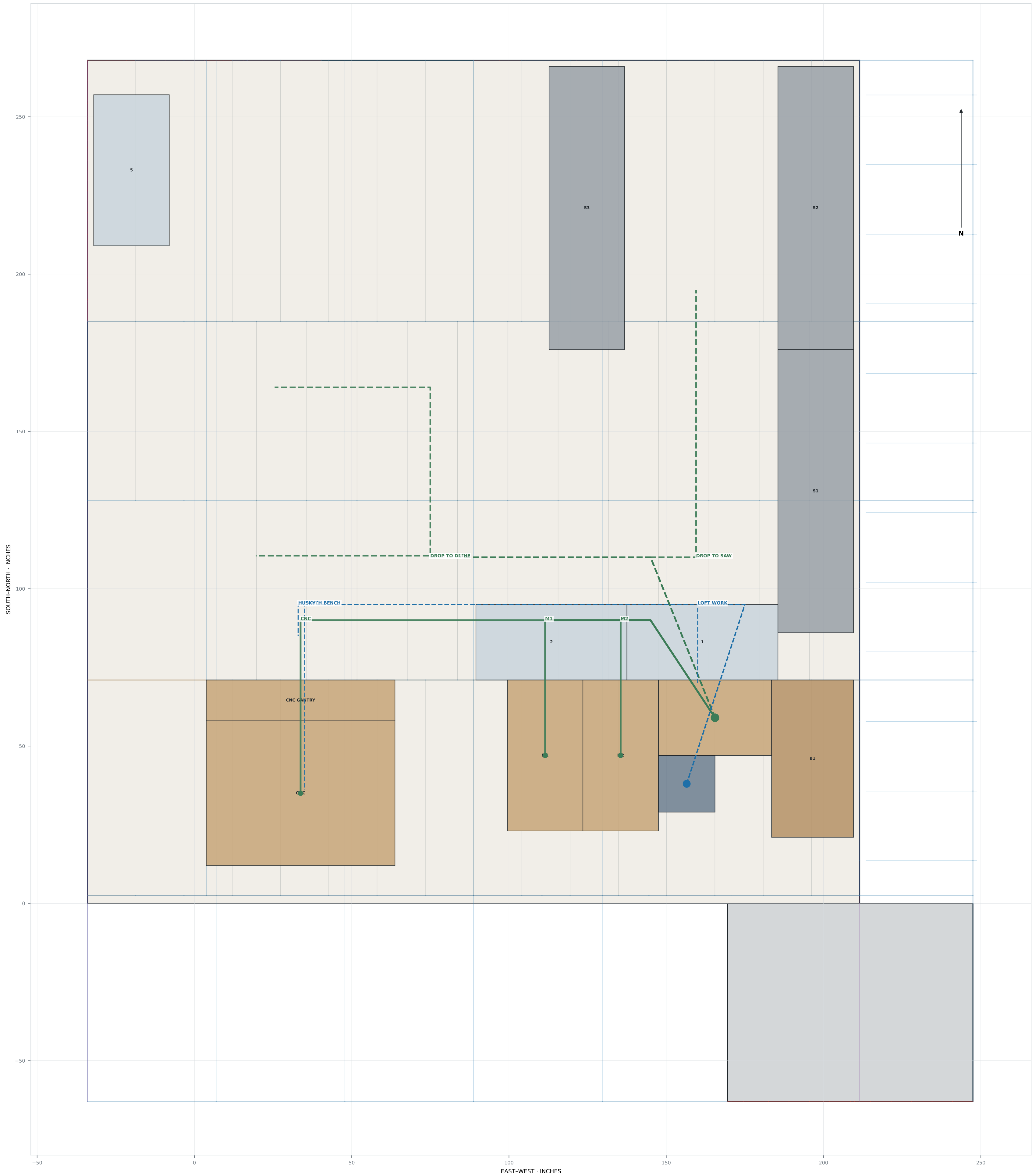
The wall-mounted 19-inch rack in the shed is the star point.  
Provide home-run cabling to shop work zones, the lathe area, loft equipment and a ceiling wireless-access-point location.  
Use pathway and spare capacity for future cameras, controls and access points.

COORDINATION NOTES

Routes are schematic; locate pathways after power and fire-protection coordination.  
Maintain separation from line-voltage conductors and sources of interference.  
Confirm cable category, patch-panel size, PoE load, rack ventilation, bonding and surge protection.  
Label both ends and record final cable IDs on an as-built schedule.

LOFT / SHOP SERVICES — CONCEPT ROUTING

SCALE: DIAGRAMMATIC



DUST COLLECTION CONCEPT

DC is the loft collector. Green lines show a conceptual main with branches to M1, M2 and CNC plus drops to the ground-floor lathe, D1 and saw. Provide a blast gate at every branch and a cleanout at low points. Final duct diameters, fan static pressure, filtration, grounding/bonding and fire protection require design from actual machine data.

COMPRESSED-AIR CONCEPT

C1 is the loft compressor. Blue dashed lines show a loop/drop concept serving the loft and both principal ground-floor benches. Slope mains to drains; provide isolation, filtration/regulation and flexible machine connections. Verify compressor ventilation, vibration isolation, relief discharge and receiver access.